

Challenges and Opportunities in Using Mixed Method Designs in Rehabilitation Research

Thilo Kroll, PhD, Jacqui Morris, PhD

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Rehabilitation research is faced with complex challenges. Heterogeneous patient populations in terms of clinical presentations, scope of functional impairment, comorbidity, and sociodemographics (eg, age, education) make powerful, blind, randomized controlled trials difficult. Even large treatment centers are often unable to provide sample sizes that would allow meaningful comparisons with controls. The problems with the feasibility of conducting randomized controlled trials are compounded by criticism directed at intervention research in terms of low external validity, and clinical and practical relevance for individual patients living in the community. This article introduces mixed method designs and discusses their relevance in the context of rehabilitation research. Examples are provided from diverse clinical research areas such as spinal cord injury and mixed disability populations.

Key Words: Methods; Qualitative research; Rehabilitation; Research design.

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THE AIM OF REHABILITATION is to support persons to maximize their physical, psychologic, and social functioning.¹ In order to achieve this, account must be taken of multifactorial and dynamic processes underpinning rehabilitation. These processes involve interactions among biological, social, psychologic, cultural, contextual, environmental, and personal factors as well as individual attitudes toward skill relearning and behavior change.² The complexity of these interactions has recently been conceptualized in the World Health Organization's ICF.³ The model, which has been embraced by the rehabilitation research community,⁴ accounts not only for disease or a health condition as a factor that influences functioning but also for contextual, personal, and environmental factors previously underrepresented in models of disability.⁵ It thus provides an integrative and dynamic biopsychosocial framework as a basis for understanding constructs and experiences of disability and processes of rehabilitation. The layers of complexity involved in rehabilitation and encapsulated in the ICF

clearly present a challenge to rehabilitation researchers wishing to investigate rehabilitation processes and their effects. The purpose of this article is to discuss some of the conceptual and methodologic challenges and to examine the contribution of mixed method designs to address the complexities inherent in rehabilitation.

The biopsychosocial approach of the ICF suggests that rehabilitation research may be seen as located at the intersection of natural and social scientific thinking.

Rehabilitation is intrinsically interdisciplinary.⁶ It combines basic science with clinical reasoning and applied research and reflects a diverse set of physical, social, cultural, and institutional conditions.⁷ It is concerned with objective and measurable outcomes of intervention practice as well as with the subjective experience of illness, injury, and recovery.

In rehabilitation research, clinical, policy, and individual service user interests may differ depending on the context of care. Interests operate in different settings (eg, hospitals, nursing homes, individual patient-provider encounters) and are expressed in terms of divergent priorities and expectations (eg, treatment adherence, social participation).

Within these contexts, different research methodologies have been traditionally assigned according to the epistemologic perspectives used by researchers in the field. Rehabilitation research has followed various design and methodological traditions, including clinical trials,⁸ survey research,⁹ and qualitative inquiry.^{10,11} Quantitative methods, primarily the experimental designs, are linked to the empirical-analytical tradition. Positivist and postpositivist researchers as proponents of this tradition are concerned with objective data, but emphasize generalizability and seek to uncover general laws between cause and effect. These quantitative designs and related methods have dominated rehabilitation science in the past because of the field's traditional connection with the objective biomedical paradigm. The biomedical model views disability as an attribute of the person, which is directly caused by disease or a health condition and which is amenable to medical intervention to correct or compensate for the problem.⁴ Thus, researchers in this field are interested in the relationship between sets of quantifiable variables and exclude questions concerning the values and intentions and personal experiences of persons—which, the empirical positivist tradition suggests, cannot be measured scientifically.¹²

Many scientists, even social scientists, have expressed considerable skepticism toward experiential knowledge. They have viewed it as prescientific and inherently biased, which is also reflected in a reluctance to publish qualitative findings.^{13,14} Qualitative research methods are linked to the historic-hermeneutic and emancipatory tradition, which is concerned with

From the School of Nursing and Midwifery, University of Dundee, Dundee, Scotland (Kroll, Morris).

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Reprint requests to Thilo Kroll, PhD, University of Dundee, Alliance for Self Care Research, School of Nursing and Midwifery, 11 Airlie Place, Dundee DD1 4HJ, Scotland, e-mail: t.kroll@dundee.ac.uk.

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List of Abbreviations

ICF	<i>International Classification of Functioning, Disability and Health</i>
RCT	randomized controlled trial

experiences, understanding, values, and meanings that persons ascribe to phenomena.¹² On the other hand, these methods are thought to produce context-sensitive, reflective information that preserves the voice of the study participant and contains snapshots of personal experiences, narratives, and interpersonal discourses over unexplored, sensitive, or complex issues. Reality is seen as a construct of social interactions and experience.¹³ With increasing emphasis on interdisciplinary biopsychosocial research in medicine, rehabilitation, and the social sciences and encapsulated in the contextual personal and environmental constructs of the ICF, acceptance of different epistemologies is needed.¹⁵ As stated by Tate, "... We make a serious mistake by reducing scientific evidence in our field [rehabilitation] to one methodological approach that mainly emphasizes quantification of data. The translation of research into practice cannot be limited by the data that is quantifiable, objective, and devoid of all individualities; on the contrary, it must include the richness of methods that can best help us to capture individual variation in its physical, social and psychological contexts. These are in essence, the gifts of our science."^{7(p165)}

Mixed method researchers adopt a pragmatic stance to incorporate experiential, context-bound knowledge into research designs that examine treatment effects between groups or relationships between predictor and outcome variables, to produce generalizable information. In this article we review key challenges in rehabilitation research and introduce mixed method design strategies.

TERMINOLOGY AND CONCEPTUALIZATION OF MIXED METHODS

A range of terms have been used for mixed method research. These include multimethod and integrative designs.¹⁶ What needs to be distinguished from the approaches presented in this article is the statistical concept of mixed model research. That pertains to a particular statistical design involving 2 types of variables—between-subjects and within-subjects variables—and not to the mixed method approaches discussed here. Mixed method designs or strategies refer to the use of both quantitative and qualitative methodologies in the context of 1 study. This design differs from separate qualitative and quantitative studies focused on the same research problem with specific research questions. The following definition captures the principal characteristics of mixed method designs:

"A mixed method study involves the collection or analysis of both quantitative and/or qualitative data in a single study in which the data are collected concurrently or sequentially"^{17(p212)} are given a priority, and involve the integration of the data at one or more stages in the process of research.

Before discussing the key elements of mixed methods, we review some of the core challenges of rehabilitation research and highlight the potential of mixed methods.

THE POTENTIAL OF MIXED METHODS IN THE CONTEXT OF REHABILITATION RESEARCH

Clinical Trial Research

Traditionally following the biomedical model, the evaluation of quality, rigor, and relevance of rehabilitation research tends to adhere to the hierarchical understanding of research evidence, which positions double-blind RCTs at its pinnacle.¹⁸ However, trials or intervention research in rehabilitation is challenged by the problem of small heterogeneous samples. A particular concern is the number of eligible study participants available at any given time. Because the time spent in inpatient

rehabilitation has been steadily declining for most long-term conditions,¹⁹ it is often not possible to enroll a large number of patients with comparable clinical characteristics and matching prognostic variables into the same study to satisfy power and sample size demands.²⁰ Funding limitations rarely allow for extended participant recruitment. Because rehabilitative care and interventions increasingly shift from inpatient to outpatient provision, it is increasingly difficult to identify or create the controlled environments needed for testing rehabilitation interventions for effectiveness. It may be impossible to apply the same standardization that would be used when conducting a pharmacological effectiveness trial. It is impossible to achieve experimental and statistical control over the complexities of environmental, intervention-related, and personal factors linked to both patients and therapists. The clinical significance, personal relevance, and external validity of findings under these conditions are therefore highly questionable if not supplemented by other methods. RCT designs have been described and criticized as being conservative in nature. They have been considered principally as designs of affirmation that offer little in terms of innovation.²¹ They enable investigators to determine whether an intervention is effective but cannot explain why it does or does not work. Complex interventions involving several components are notoriously difficult to evaluate because of problems in developing, identifying, documenting, and reproducing the intervention.²⁰ Moreover, many rehabilitation intervention trials have been reported incompletely in the literature, which makes conclusions about the state of the evidence challenging.²²

In the light of these challenges, much of the rehabilitation research literature relating to intervention effectiveness is not based on RCTs. Many studies use single case designs²³ or preintervention and postintervention designs,²⁴ thereby failing to reach the highest level of the evidence hierarchy.²⁵ In the area of intervention research, mixed method research may thus contribute to the understanding of failed and successful rehabilitation treatments and the acceptability of interventions, and may help explain unintended or serendipitous treatment effects. Mixed method designs that combine exploratory and confirmatory research components may be superior to traditional RCT models when applied to rehabilitation research because they go beyond simple hypothesis testing toward generation of insights into the mechanisms that may be responsible for differences in outcomes between groups. These designs can allow researchers to examine critical influencing factors and variables before studying the effectiveness of interventions. There are examples from primary care and several community-based interventions of trial designs that use qualitative methods before, during, or after an intervention trial.²⁶ These approaches enable researchers to examine outcomes of trials in greater depth by identifying additional benefits reported by study participants, an evaluation not typically included as part of the trial methodology.^{27,28}

Outcome Measures

The identification of appropriate outcome measures and instruments is another major challenge for rehabilitation researchers. Generic tools, such as the Medical Outcomes Study 36-Item Short-Form Health Survey,²⁹ that are used to assess quality of life are relatively superficial measures and potentially meaningless to persons who have specific rehabilitative goals linked to life roles, social activities, and societal participation. The use of generic measures is unspecific and limited in explanatory value. Change in the measure may have little relevance for persons who may have completed a vocational rehabilitation program to enhance their chances for a return to

work after stroke. Measures that are not sensitive to changes in functional capacity may fail to capture effects of successful rehabilitation. This is of particular relevance in conditions such as multiple sclerosis and brain injury rehabilitation, in which status may deteriorate or become static over time, but improvements in activity limitation or participation restriction can and do occur.³⁰ Tools such as the Barthel Index³¹ or the FIM³² do not reflect how the individual adapts or copes with the challenges associated with the management of long-term disabling conditions.

To date, there are few validated tools in existence that have been tested for specific disability populations (eg, participation measures for people after traumatic brain injury), although developments are under way.³³ Moreover, many measures either are paper-and-pencil-based and are not suitable for people with limited hand control, or rely on proxy respondents and are removed from the experience of the individual patient.

Mixed method research may contribute to the development of relevant outcome measures. Firstly, exploratory qualitative components can inform researchers and practitioners about what outcomes are most meaningful and relevant for patient subpopulations as well as about the preferred and most useful format for collecting information. In post hoc evaluations of intervention effects, qualitative examination can identify why differences in intervention outcomes occurred, uncover additional treatment benefits, and explore barriers to achieving the best intervention outcomes.

Exclusion From Research

People with disabilities have traditionally faced physical and social exclusion from agenda setting in rehabilitation research.^{34,35} It is not surprising that consequently, many disability advocates are highly critical of, skeptical of, or even opposed to rehabilitation research. For example, physical exclusion from research is occurring in experimental clinical studies because of (1) tight inclusion criteria (eg, participation in the study requires a pass in the Mini-Mental State Examination) and (2) inappropriate instrumentation (eg, alternative formats such as braille version of a questionnaire is not made available; the lack of a wheelchair accessible scale does not allow for weight measurement; sign language interpreters have not been budgeted).

Social exclusion from research may come in many shapes or forms. Researchers may not be aware or they do not look in the direction of the excluded.³⁶ Some may refer to the fact that study budgets are too small to allow for adaptations in the research process that would enable persons requiring alternative formats to participate. Investigators may be insufficiently trained to communicate effectively with people with disabilities, especially with persons who have complex impairments involving cognition and speech. Researchers may not understand or may not be motivated to address the specific augmentative communication and assistance needs.³⁷

As a reason for not yielding to the requirements for adaptations in the research process, some researchers may cite that they are concerned about compromising the integrity of their research protocols if they stray too far from standardized approaches. However, the point that their findings would be strictly limited to those who meet their strict inclusion criteria is rarely made when findings and implications are discussed. The blanket assumption is made that results would apply equally to those who have not been asked. Many researchers use validated or standardized measures that actually lack validation for particular subpopulations or compare findings with norms that are not appropriate.

The issue of exclusion is important not only from an ethical or social justice perspective but also in terms of quality assurance in medical and social research, particularly with regard to validity and generalizability of findings and applicability in rehabilitative practice. Exclusion skews the evidence base. This in turn may affect resource allocation in health and social care systems and wider social policies. Mixed method research by way of constructively combining qualitative and quantitative study components and measures may improve not only the reach of rehabilitation research but also the fit between research approach or method and research participants. It may help address exclusion biases and enhance research participation of traditionally excluded groups and make their needs, experiences, and perspectives visible.

Planning Mixed Method Studies

Apart from scientific appropriateness, mixed method research requires specific expertise, funding, and time. Funding bodies are still in the process of developing expertise in the area of mixed method research, and investigators may encounter reluctance to fund research involving these designs. However, there is a growing interest among funders to follow a roadmap that links basic with contextual research and applied effectiveness evaluations of clinical interventions.³⁸ In addition to presenting a scientific rationale for the selection of a mixed method design strategy, investigators need to demonstrate that they have the necessary expertise to conduct the study in the time and with the budget available to them. Principal decisions need to be made about how quantitative and qualitative components are sequenced, prioritized, and integrated.³⁹

Mixed Method Design Types, Sampling, and Methods

There is a considerable variety of mixed method designs described in the literature,⁴⁰ but these can be reduced to a few core designs with discernible signature features.

Implementation Sequence: Concurrent and Sequential Designs

The sequence of implementation is important. It relates to the decision about how qualitative and quantitative study components are arranged. The fundamental question is whether qualitative and quantitative methods will be used concurrently or sequentially. In sequential designs, qualitative and quantitative data collection phases follow one another. In concurrent studies, both types of data are collected at the same time.

For example, in a health promotion research study, focus groups were used with persons with spinal cord injury to identify barriers to exercise, preferred exercise formats, and the potential role and function of peer support.⁴¹ The focus group findings were linked to steps in the development of a multi-component exercise program, which was evaluated using a controlled evaluation approach and standardized scales for outcome measurement. We used a sequential design approach in which the qualitative study component was followed by a quantitative evaluation. Both study components were linked through the purpose of the research study, which was to identify facilitators of effective exercise in people with spinal cord injury.

In a second example of a sequential study, focus groups (qualitative study component) were used to explore the health care access experiences of homeless adults with disabilities.⁴² The information obtained informed the design of a survey instrument (quantitative evaluation) that encompassed the critical areas highlighted in the focus groups, which was then fielded with a larger group of homeless adults with disabilities in the District of Columbia.⁴³

The 2 examples describe the most common sequence of how qualitative and quantitative methods are being combined: a qualitative study component followed by quantitative evaluation. The rationale for this sequence is that a new or unfamiliar area of research or a new intervention setting is first explored qualitatively to identify key factors that may influence outcomes. After the initial exploration, these observations are tested in a larger sample. Here the size of the effect or the frequency of occurrence of a phenomenon and the statistical associations among variables are described. However, the reverse sequence is also found. For example, if investigators are interested in finding possible explanations for survey results or outcomes from an experimental evaluation of different treatments, they may follow up a particularly interesting subset of study participants qualitatively (quantitative evaluation → qualitative study component).⁴⁴ This is a study in which quantitative evaluation of an exercise class in which no significant effects were found was followed up by a qualitative examination. The qualitative analysis showed important physical, social, and psychologic improvements undetected by the quantitative evaluation.

In a prospective study that was conducted to examine whether activity status and exercise behavior patterns were linked to the onset of certain medical complications (eg, pressure ulcers, urinary tract infections, respiratory complications) and secondary conditions (eg, osteoporosis), qualitative in-depth telephone interviews (qualitative study component) were conducted with a purposefully selected subset of respondents concurrently with the ongoing quantitative survey study. The interviews provided an in-depth examination of some of the associations found in the first wave of the survey (quantitative evaluation). Participants were asked about the type of exercises and physical activities they were engaged in, their exercise history, and their perceptions about possible links between exercise behavior and the prevention of secondary medical problems.⁴⁵

Priority

Priority refers to which study component takes precedence or is of greater importance than the other. It refers to the relative weight assigned to the qualitative and quantitative research components.⁴⁶ The dominance or priority of the research components can be illustrated by using capital letters in the abbreviation of quantitative or qualitative components. The use of capitals for both indicates that equal priority is assigned to each method, whereas use of lower case indicates lower assigned priority within the study. In areas with little prior evidence, qualitative methods may be used to determine the scope and nature of concepts and variables, and the relationships among them. In these circumstances, greater priority may be assigned to the qualitative study component because this work is absolutely essential before the key variables can be studied quantitatively (QUAL-*quan*). Conversely, in explanatory research, in which qualitative research is used mostly to substantiate findings generated through quantitative surveys or experiments, priority is usually assigned to the quantitative component (QUAN-*qual*).

Integration

A further key feature of mixed methods is purposeful integration. Integration can occur at different stages of the research process. The decision about where integration occurs and in which way is linked to the research question and the aforementioned motivation of connected contributions. Researchers need to have a clear idea about how the different data compo-

nents inform one another and how they provide distinctive answers to the research questions derived from the overall research purpose.^{32,38} Integration can occur at the point the data are collected (eg, combined survey and qualitative in-depth interview; cognitive testing interviews of a newly developed scale) or later, during data analysis (eg, cross-tabulations of interview codes are linked back to survey variables such as demographics) and interpretation (eg, convergence, divergence, complementarity of findings obtained with different methods). Software has been developed that increasingly eases the burden of handling both qualitative and quantitative data through improved import and export from statistical into qualitative data analysis packages and vice versa.^{47,48}

Theoretical Perspective

A particular theoretical stance or orientation may influence the selection of a particular mixed method design strategy and the research process as a whole.⁴⁹ Because participatory research is increasingly emphasized by funding bodies, researchers may strive to maximize involvement of key stakeholders at all stages of the process.⁵⁰ Action researchers who rely on mixed method approaches may turn to so-called transformative designs. These design strategies emphasize change in organizational or health care management practice as a result of the research and a collaborative ethos between researchers and key stakeholders.⁵¹ Theory may also underpin mixed method research in other ways. The study may be guided by formalized empirical theories (eg, social cognitive theory),⁵² epistemologic positions (eg, phenomenology),⁵³ or social theories (eg, social model of disability).⁵⁴

Strengths and Weaknesses of Mixed Method Designs

Mixed method designs clearly have theoretical and practical strengths in terms of (1) greater stakeholder involvement in the research process, (2) enhanced context sensitivity in line with models such as the ICF leading to potential ecological and external validity of research, and (3) enhanced exploratory-confirmatory linkages or road mapping from modeling through confirmatory trial research. However, some weaknesses should be noted here as well. Mixed method research is costly and time-consuming and requires specialist expertise in a range of methods. At present, mixed methods remain widely untested in the field of rehabilitation research. Some research is published under the label mixed methods as an afterthought rather than as a planned mixed method research study. Conversely, studies that could be classified as mixed methods are published in a fragmented way—that is, qualitative findings and quantitative results are reported in separate articles. The nomenclature and use of mixed method design strategies continues to be developed, and disagreement can be found among theorists about the direction of mixed method terminology (eg, transformative vs pragmatic), and what should be considered mixed method research and what should not.

This ongoing debate may have left rehabilitation researchers with a sense of uneasiness and reluctance in terms of embracing mixed methods fully in their canon of design strategies.

CONCLUSIONS

Mixed method designs are new developments in the context of rehabilitation research, and their full potential has yet to be explored. In an era of evidence-based medicine and practice,⁵⁵ reliance on the conduct of RCTs as the principle source of evidence has been met with criticism as limitations have become visible. Such a narrow view of evidence is an impossibility for community-based or even clinical rehabilitation re-

search. There is mounting discontent in practice and policy communities about the delays in translating research findings into practice. Moreover, service users often question the relevance and utility of clinical research for their lives. The United Kingdom's Medical Research Council framework for the design and evaluation of complex interventions embeds trial research in an iterative process of exploratory research, modeling, and feasibility testing. This approach allows researchers to scrutinize the implementation context, the barriers and facilitators that may hinder or promote the efficacy of interventions, before costly trials are conducted.⁵⁶ Mixed method research clearly plays an important role within this framework, because exploratory, explanatory, and effectiveness components form an iterative process that may enhance the utility and relevance of rehabilitation results in different clinical and community settings. Mixed method research is also associated with greater potential for inclusion of key stakeholders, especially people with disabilities. Active service user participation increases the relevance of findings for their lives and challenges researchers to create a study environment that maximizes inclusion through the use of flexible and alternative formats (especially in the context of paper-and-pencil dominated research). Epistemologically, mixed methods may provide an arsenal of pragmatic approaches with a view of broadening the concept of knowledge and enhancing translational efforts among service users with disabilities, rehabilitation researchers, clinical practitioners, and policy makers.

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